Research Paper Citation Integration Guide

Placement, Natural In-Text Phrasing, and References for IEEE Paper (SecureKYC AI)

Target Manuscript: `paper/ieee_research_paper.md`
Target Format: IEEE Conference / Scopus Indexed Target
Google Scholar Profile: Prof. (Dr.) Rajesh Kumar Dhanaraj
Corpus Examined: All 880 publications on profile

1. THE THREE SELECTED PAPERS

From the profile's full publication list, these three papers connect directly to the technical building blocks of your system: multimodal facial verification, continuous authentication across video sessions, and facial landmark tracking under real-world webcam conditions.

Rank	Paper Title	Authors	Venue & Year	Role in Your Paper
RANK 1	**Secure facial biometric authentication in smart cities using multimodal methodology**	A. Sureshkumar, M. Sathyamoorthy, R. K. Dhanaraj, S. Aanjanadevi, et al.	*Scientific Reports*, 2025	Section I (Introduction): Motivates why multimodal checks are necessary over single-frame facial recognition.
RANK 2	**A comprehensive study on continuous person authentication using behavioral biometrics**	A. B. Badade, R. K. Dhanaraj	*IEEE TQET*, 2024	Section I (The Puppeteer Problem): Explains why static checks fail and why verification must be continuous throughout the call.
RANK 3	**Facial Expression Recognition in the Wild using Artificial Rabbits Optimizer based Residual Neural Network**	P. S. Metkewar, N. Uma, R. K. Dhanaraj, B. P. Kavin, M. Sathyamoorthy	*IEEE ICAICCIT*, 2023	Section III-A (Lip Tracking): Justifies normalizing 3D landmarks against unconstrained webcam lighting and head tilt.

2. WHERE TO ADD EACH CITATION (CLEAN, NATURAL PHRASING)

Paper 1: Sureshkumar et al. (2025) — Multimodal Facial Biometrics

CITATION [22]

A. Sureshkumar, M. Sathyamoorthy, R. K. Dhanaraj, S. Aanjanadevi, and P. Chatterjee

Scientific Reports, vol. 15, no. 1, art. 44839, 2025

Location: Section I. INTRODUCTION (Paragraph 1, lines 26–27)

Context: Place this where you introduce remote video KYC. It shows that current systems combine multiple biometric signals to improve accuracy, but still rely on the assumption that a physical camera is present.

– CURRENT TEXT:

"Onboarding of remote users in accordance with RBI Master Directions on KYC and Europe's AMLD5 and eIDAS rules requires a WebRTC video call session. The paradigm assumes a live applicant positioned in front of an optical camera sensor. However, this assumption is negated by modern real-time generative models. Traditional presentation attacks... required traversal of the optical path. In the case of today's attacks, synthetic frames can be injected directly into the input stream [1], [2]."

+ NATURAL REPLACEMENT (WITH CITATION [22]):

"Onboarding of remote users in accordance with RBI Master Directions on KYC and Europe's AMLD5 and eIDAS rules requires a WebRTC video call session. While recent verification schemes combine multimodal biometric cues to improve authentication reliability [22], current protocols still assume the applicant is physically in front of an optical sensor. Modern generative tools break this assumption completely: instead of pointing a photo or screen at a webcam, attackers can feed synthetic frames directly into the browser video stream [1], [2]."

Paper 2: Badade & Dhanaraj (2024) — Continuous Person Authentication

CITATION [23]

A. B. Badade and R. K. Dhanaraj

2024 IEEE International Conference on Trends in Quantum Computing and Emerging Technologies (TQET)

Location: Section I. INTRODUCTION (Paragraph 3: The Puppeteer Problem, lines 30–31)

Context: Place this where you explain why passive checks fail. An attacker can pass a one-time check and then start the deepfake feed; this citation shows why authentication must run continuously across the entire interaction.

– CURRENT TEXT:

"The real-time puppetry makes the injection interactive. The attacker sits at a webcam and speaks, and a reenactment tool such as DeepFaceLive (built on DeepFaceLab) tracks their facial keypoints frame by frame and renders the victim's face over those movements [8]. Passive presentation attack detection (PAD) in the sense of ISO/IEC 30107-3 does not stop this. These blink, smile, and yaw triggers fail since the attacker carries out the trigger, and the neural mask simulates it with sub-frame latency."

+ NATURAL REPLACEMENT (WITH CITATION [23]):

"Real-time puppetry makes injection attacks interactive. An attacker speaks into their own webcam, and a tool like DeepFaceLive tracks their face and maps the victim's identity onto those movements frame by frame [8]. Passive presentation attack detection cannot stop this. Single-action checks verify only an isolated moment, leaving the rest of the call unprotected; while continuous authentication using behavioral biometrics has been proposed to prevent mid-call session hijacking [23], simple triggers like blinking or smiling still fail here because the human attacker simply carries out the motion, which the model mirrors almost instantaneously."

Paper 3: Metkewar et al. (2023) — In-the-Wild Facial Feature Extraction

CITATION [24]

P. S. Metkewar, N. Uma, R. K. Dhanaraj, B. P. Kavin, and M. Sathyamoorthy

2023 IEEE International Conference on Advances in Computation, Communication and Information Technology (ICAICCIT)

Location: Section III-A. Real-Time 3D Lip Aperture Metric (lines 95–96)

Context: Place this where you introduce MediaPipe Face Mesh tracking and the normalized aperture formula. It justifies why normalizing the aperture is necessary to handle unconstrained, low-cost consumer webcams.

– CURRENT TEXT:

"The lip aperture is computed independently of camera distance by tracking 3D facial landmarks with the MediaPipe Face Mesh [15] library, with landmark refinement enabled and the face-detection confidence threshold set to 0.5. Let p_i = (x_i, y_i, z_i) be the coordinates of the i-th landmark..."

+ NATURAL REPLACEMENT (WITH CITATION [24]):

"We track 3D facial landmarks using MediaPipe Face Mesh [15] with landmark refinement enabled and a 0.5 detection threshold. In consumer video calls, facial analysis must handle unconstrained conditions like poor lighting, head turning, and rapid expression changes during speech [24]. To keep the lip aperture metric independent of camera distance and scale, we normalize the vertical lip separation by the subject's face height (distance between nose tip and chin). Let p_i = (x_i, y_i, z_i) be the coordinate of landmark i:"

3. STANDARD IEEE REFERENCES TO APPEND

Paste these three entries at the end of the `## REFERENCES` section in `paper/ieee_research_paper.md`:

> [22] A. Sureshkumar, M. Sathyamoorthy, R. K. Dhanaraj, S. Aanjanadevi, and P. Chatterjee, "Secure facial biometric authentication in smart cities using multimodal methodology," *Scientific Reports*, vol. 15, no. 1, art. no. 44839, 2025.
>
> [23] A. B. Badade and R. K. Dhanaraj, "A comprehensive study on continuous person authentication using behavioral biometrics," in *Proc. IEEE Int. Conf. Trends Quantum Comput. Emerg. Technol. (TQET)*, 2024, pp. 1–6.
>
> [24] P. S. Metkewar, N. Uma, R. K. Dhanaraj, B. P. Kavin, and M. Sathyamoorthy, "Facial Expression Recognition in the Wild using Artificial Rabbits Optimizer based Residual Neural Network," in *Proc. IEEE Int. Conf. Adv. Comput., Commun. Inf. Technol. (ICAICCIT)*, 2023, pp. 248–253.

4. REVIEWER DEFENSE NOTES

If reviewers ask about these topics during submission, these three citations give you solid grounds to answer:

Possible Reviewer Question	Cited Reference	Your Defense in the Text
Why use multimodal fusion instead of just a fine-tuned CNN on face images?	**Sureshkumar et al. (2025) [22]**	Single-image face models are easy to fool with virtual camera feeds; multimodal biometric verification is needed to catch inconsistencies across modalities.
Why not just use a single passive selfie check at the start of the call?	**Badade & Dhanaraj (2024) [23]**	A single-point check leaves the rest of the call unprotected; verification needs to be continuous across the whole session to prevent feed swapping.
How does your lip tracker handle cheap webcams, bad lighting, or tilted heads?	**Metkewar et al. (2023) [24]**	Real-world webcam conditions are noisy; normalizing lip distance by face height prevents scale errors when the user moves or tilts their head.